

Ally's Stream Hub

24/7 Free Cloud Hosting Guide (Oracle Cloud Always Free)

Host your TikTok Live game suite online with zero server fees, SSL, and PM2.

Overview & Prerequisites

Host Ally's Stream Hub on Oracle Cloud Infrastructure (OCI) using their Always Free tier. Your server runs 24/7 online, accessible by OBS and mobile browsers anywhere in the world without keeping your personal computer running.

- **Oracle Cloud Account:** An active OCI account (credit card verified during signup, but \$0 billed).
- **SSH Client:** PowerShell (built into Windows 10/11) or Windows Terminal.
- **Git Repository:** Your code at <https://github.com/MrDDubs/odd-one.git>

Step 1: Create Free Compute Instance in Oracle Cloud

1. Sign in to your Oracle Cloud Console at <https://cloud.oracle.com/>
2. From the navigation menu (top-left hamburger), select Compute -> Instances.
3. Click Create Instance and configure the following parameters:
 - **Name:** ally-stream-hub
 - **Image:** Click 'Edit' -> select Canonical Ubuntu -> choose Ubuntu 22.04 or 24.04.
 - **Shape:** Ampere ARM (VM.Standard.A1.Flex, 2 OCPUs, 12 GB RAM) - Always Free eligible.
(If ARM capacity is temporarily full in your region, choose AMD VM.Standard.E2.1.Micro, 1 GB RAM).
 - **Networking:** Leave default VCN. Confirm 'Assign a public IPv4 address' is checked.
 - **SSH Keys:** Select 'Generate a key pair for me' -> click 'Save Private Key'. Store this .key file safely on your PC (e.g. C:\Users\Dwayne\oci_key.key).
4. Click Create at the bottom. Wait 1-2 minutes until status turns green (Running).
5. Copy your Public IP Address shown on the instance screen (e.g. 140.238.xx.xx).

Step 2: Open Ports in Oracle Cloud Network (VCN Security List)

Oracle Cloud blocks incoming traffic by default. Open ports 80 (HTTP), 443 (HTTPS), and 4000 (Stream Hub):

1. On the Instance page, scroll down to Primary VNIC and click the Subnet link.
2. Under Security Lists, click on Default Security List for...
3. Click Add Ingress Rules and fill in:
 - **Source CIDR:** 0.0.0.0/0
 - **IP Protocol:** TCP
 - **Destination Port Range:** 80, 443, 4000
 - **Description:** Allow Web, HTTPS, and Stream Hub traffic
4. Click Add Ingress Rules.

Step 3: Connect to your Server via SSH

Open PowerShell on your Windows PC and run:

```
ssh -i "C:\path\to\your_saved_key.key" ubuntu@YOUR_PUBLIC_IP
```

Type 'yes' when prompted to accept the host key. You are now inside your cloud Ubuntu server!

Step 4: Configure Ubuntu Firewall & Install Dependencies

CRITICAL ORACLE LINUX STEP

Ubuntu images in Oracle Cloud include restrictive internal iptables rules that drop ports even after opening them in the web console. You must run the commands below to allow traffic through.

Run these commands inside your SSH terminal:

```
# 1. Update system packages
sudo apt update && sudo apt upgrade -y

# 2. Open ports in Ubuntu's internal firewall
sudo iptables -I INPUT 6 -m state --state NEW -p tcp --dport 80 -j ACCEPT
sudo iptables -I INPUT 6 -m state --state NEW -p tcp --dport 443 -j ACCEPT
sudo iptables -I INPUT 6 -m state --state NEW -p tcp --dport 4000 -j ACCEPT
sudo apt install -y iptables-persistent
sudo netfilter-persistent save

# 3. Install Node.js 22 LTS, Git, and Nginx
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt install -y nodejs git nginx

# 4. Install PM2 (Process Manager to keep app running 24/7)
sudo npm install -g pm2
```

Step 5: Clone and Launch Ally's Stream Hub

Clone your repository from GitHub and install production dependencies:

```
# Clone repository
git clone https://github.com/MrDDubs/odd-one.git stream-hub
cd stream-hub

# Install production dependencies
npm install --omit=dev

# Launch server with PM2
pm2 start server.js --name "stream-hub"

# Configure PM2 to auto-restart on system reboot
pm2 save
pm2 startup
```

Note: If 'pm2 startup' displays a command starting with 'sudo env PATH=...', copy and run it.

Step 6: Set Up Nginx Reverse Proxy with WebSocket Support

Nginx routes incoming web and WebSocket traffic directly into your Stream Hub Node.js process without needing to specify port :4000 in your URLs.

1. Create an Nginx site configuration: `sudo nano /etc/nginx/sites-available/stream-hub`
2. Paste this configuration block:

```
server {
    listen 80;
    server_name _;

    location / {
        proxy_pass http://localhost:4000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}
```

Save and exit: Press Ctrl+O, Enter, then Ctrl+X.

3. Enable the site and restart Nginx:

```
sudo ln -s /etc/nginx/sites-available/stream-hub /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t && sudo systemctl restart nginx
```

Step 7: Accessing Your Live Stream Hub Online

Your server is live online 24/7! You can access all endpoints immediately using your Public IP:

Destination	URL	Usage
OBS Stream Overlay	<code>http://YOUR_PUBLIC_IP/overlay.html</code>	OBS Browser Source (1080x1920)
Host Controls	<code>http://YOUR_PUBLIC_IP/admin-overlay.html</code>	Phone, Tablet, or PC Control Room
Hub Home / Launcher	<code>http://YOUR_PUBLIC_IP/</code>	Game Selector & Status
Raw Node Port (Optional)	<code>http://YOUR_PUBLIC_IP:4000/</code>	Direct Socket.IO & API Port

Step 8: Optional: Free Domain & HTTPS/SSL (Let's Encrypt)

Modern browsers and OBS overlays often require HTTPS / WSS for camera, secure assets, or permissions. You can configure a completely free custom domain and SSL certificate:

1. Get a free domain or subdomain (e.g., at DuckDNS.org or Cloudflare pointing to your Public IP).
2. Update /etc/nginx/sites-available/stream-hub and change 'server_name _;' to 'server_name yourname.duckdns.org;'.
3. Install Certbot and generate an automated SSL certificate:

```
sudo apt install -y certbot python3-certbot-nginx
sudo certbot --nginx -d yourname.duckdns.org
```

Certbot automatically handles certificate renewals forever via background systemd timers.

Step 9: Maintenance & Quick Update Cheat Sheet

UPDATING YOUR HUB IN 10 SECONDS

Whenever you push code updates or new games to GitHub, SSH into your server and run:
`cd ~/stream-hub && git pull origin master && pm2 restart stream-hub`

Useful PM2 diagnostic commands:

```
# View live logs in real time
pm2 logs stream-hub

# View CPU and RAM usage dashboard
pm2 monit

# Restart or stop the hub
pm2 restart stream-hub
pm2 stop stream-hub

# Check status and uptime
pm2 status
```